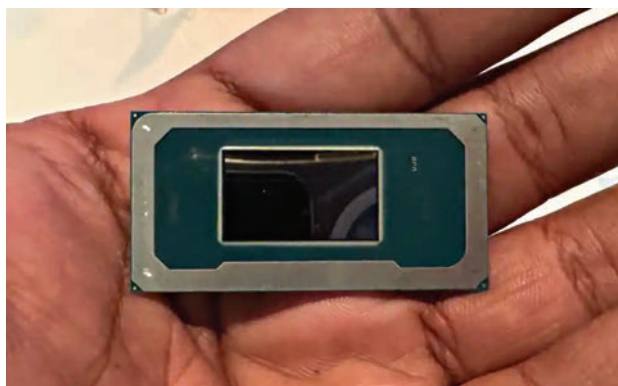




Intel unveils Panther Lake: First AI PC chip built on 18A

Intel took the wraps off its third-generation of AI PC chip, Intel Core Ultra series 3 – codenamed Panther Lake. It's Intel's next processor update to feature in laptops starting in 2026, and the first to be built on its new 18A semiconductor process. Set to power a wide range of AI-enabled PCs and edge devices, Intel Core Ultra series 3 or Panther Lake chips will begin shipping later this year, with high-volume production ramping at Intel's newest manufacturing facility, Fab 52, in Chandler, Arizona.

Intel also offered an early look at Clearwater Forest – its first Xeon server processor based on 18A – scheduled to launch in the first half of 2026. These announcements were made as part of the company's 2025 Tech Tour, which Digit attended, marking a key milestone in Intel's multi-year roadmap to regain leadership in process technology and semiconductor manufacturing. Both chips are being produced at Fab 52, Intel's most advanced factory to date and part of its \$100 billion US expansion strategy.

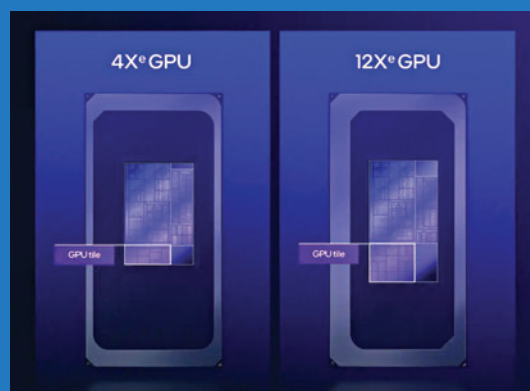
Compared to TSMC's 3nm (N3) and upcoming 2nm (N2) nodes, Intel 18A introduces two defining innovations: RibbonFET, Intel's gate-all-around transistor design (its first new transistor architecture in over a decade), and PowerVia, a backside power delivery system that routes power through the rear of the chip – freeing up the front for improved signal routing and reducing voltage droop.

Intel claims these combined advances deliver up to 15% better performance per watt and 30% higher chip density compared to Intel 3, putting 18A on competitive, if not leading, footing with TSMC's N2 process in terms of transistor technology. Crucially, 18A is also being manufactured domestically in the United States., giving it geopolitical and supply-chain relevance that TSMC – still reliant on Taiwan for leading-edge production – currently lacks, according to Intel.

Panther Lake's Xe3 iGPU could game and do AI better

It's because Intel's Panther Lake platform is built around a much beefier integrated GPU and a software stack that leans hard into AI-assisted rendering. According to initial benchmarking claims from the company, 2026 Intel laptops should deliver playable 1080p – and often much higher – on modern titles, while also accelerating creator and AI workflows without a discrete GPU.

Intel's next-gen upscaler, XeSS 3, adds Multi-Frame Generation (MFG) that can synthesize up to three AI frames between real frames, potentially multiplying perceived frame rate two to four times depending on settings and content. Crucially, Intel says XeSS-MFG isn't locked to just the very latest silicon – the company is signaling



broader compatibility reaching back to earlier Arc and integrated generations, which bodes well for developers targeting a wide base – and for buyers who don't want instant obsolescence of their previous-gen Intel laptops.

Panther Lake's Xe3 graphics architecture is the muscle behind the above claims, with reports pointing to 50%+ gains over last gen and features on par with discrete-class GPUs. Expect ray tracing support, updated media engines, and driver-level controls that make AI features easier to toggle per game.

On the platform side, Microsoft is preparing DirectX upgrades aimed at making these AI paths first-class citizens in the graphics pipeline. That alignment is key because then games and creative apps can tap standard APIs more easily.